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Using digital mobile games to increase the support for nature conservation

Diogo Veríssimo[1] | Sicily Fiennes[2] | Matilda Dunn[1,3]

[1]Department of Biology, University of Oxford, Oxford, UK
[2]School of Biology, Faculty of Biological Sciences, University of Leeds, Leeds, UK
[3]Faculty of Natural Sciences, Centre for Environmental Policy, Imperial College London, London, UK

Correspondence
Diogo Veríssimo, Department of Biology, University of Oxford, 11a Mansfield Road, Oxford OX1 3SZ, UK.
Email: diogo.gasparverissimo@biology.ox.ac.uk

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Abstract

Digital games are an increasingly dominant form of digital entertainment with billions of players globally. While most of these games have a commercial focus, fields like public health and education have seen a growth of "serious games," which aim to solve real world problems. In the context of biodiversity conservation, mobile games have been controversial, with some raising concerns around the way digital channels risk replacing the very nature they portray, therefore deepening a "nature-deficit disorder." We ran a randomized controlled trial to evaluate the impact of the mobile game "Kākāpō Run" on pro-environmental behaviors amongst a cohort of 200 participants in New Zealand. Kākāpō Run was developed by a UK conservation charity, and aims to increase the support for Kākāpō conservation, as well as to increase pro-environmental behaviors linked to Kākāpō conservation. Study participants completed a 10-minute questionnaire before spending 1 hour playing their assigned mobile game over seven days. This was monitored by asking participants to share screenshots of their app usage for the duration of the experiment. After this, all participants re-took the questionnaire. We found a positive impact across some knowledge and attitudes questions, behavioral intentions linked to willingness to volunteer time and support policies aiming to remove invasive predators, as well as manage pet cats actively. However, we found no change in willingness to donate or actual donations. This research showcases both the potential of mobile games for conservation outreach and marketing, and the importance of rigorous impact evaluation. We call for conservationists engaged in designing and promoting mobile games to approach game design and evaluation in a more research-centered way to help develop an evidence base around the intended and unintended consequences of game playing. This mainstreaming of gaming science across conservation will be critical to allow mobile games to realize their potential as a leading communication channel.

KEYWORDS
behavior change, digital, education, online, outreach, serious games, storytelling

1 | INTRODUCTION

Digital games have become a dominant form of entertainment globally, a trend that has only accelerated further during the COVID-19 pandemic (Deloitte Insights, 2020). Within digital games, those played on mobile devices (i.e., smartphones or tablets) now constitute the largest market segment, with an expected 3 billion gamers in 2023 and projected revenues of $77.2 billion, nearly half of the total digital game revenue (Newzoo, 2020). Thus, digital games are no longer seen as a strictly juvenile pastime (Colleton et al., 2016), having hit the global mainstream, with Africa and Middle East having a similar number of players to Europe, and Latin America having more players than North America (Newzoo, 2020).

While most digital games are commercial in nature, the last decade has seen a growth in so called "serious games" which hope to contribute to solving real world problems (Annetta, 2010; Sandbrook et al., 2015). Mobile app games are an increasingly appealing method of public engagement for achieving a social or "common" good (McGonigal, 2010; Newzoo, 2020). This is unsurprising given the body of evidence that suggests they can be an effective tool to improve multiple types of outcomes, from knowledge acquisition and cognitive skills, motivational and behavioral change (Boyle et al., 2016; Clark et al., 2016), as well as across multiple topic areas including those of particular policy concern, such as healthy eating, sexual health, language training, and physical fitness (Boyle et al., 2016).

In the context of biodiversity conservation, while the potential for games to benefit biodiversity has been long recognized, there has also historically been opposition to their inclusion in the portfolio of outreach methods used by conservationists (Fletcher, 2017; Greenwood, 2012; Sandbrook et al., 2015). Detractors of digital games argue these games risk replacing the very nature they portray, deepening a "nature-deficit disorder" (Louv, 2008) and luring players into a false sense of security about the state of the natural world (Arts et al., 2015; Fletcher, 2017). Proponents of digital games argue that games can reach audiences that are beyond traditional outreach approaches. They argue that to ignore a leading form of entertainment globally is a crucial mistake at a time when conservation is in dire need of outreach solutions that can be globally scalable (Annetta, 2010; Greenwood, 2012; McGonigal, 2010; Thomas-Walters & Veríssimo, 2022). This debate has unfortunately not substantially evolved in the last decade. While Fletcher (2017) and Sandbrook et al. (2015) offered numerous credible pathways when it comes to the ability of digital games to positively impact human behaviors relevant to biodiversity conservation, there has been little research into the outcomes of the multiple games that have been launched into this time period (see Colleton et al., 2016; Phipps et al., 2016; Salvador & Cunha, 2016), with exceptions being Dunn et al. (2021) and Thomas-Walters and Veríssimo (2022).

Our research aims to move this discussion forward by rigorously assessing the impact of a conservation focused mobile app games on players' pro-environmental behaviors when compared to a similar game with no conservation messaging. We tested the conservation game "Kākāpō Run," a mobile app game which aims to inform players about the threats faced by the native New Zealand bird, the Kākāpō, through the format of a runner mobile game (Figure 1). The Kākāpō bird (*Strigops habroptilus*), has been classed as Critically Endangered by the IUCN since 2000 (BirdLife International, 2000). One of the most significant threats faced by the bird is from invasive non-native species to the New Zealand archipelago, including domestic cats, stoats (*Mustela erminea*) and black rats (*Rattus rattus*) (Clout & Merton, 1998; Collar & Sharpe, 2014). The objective of the game is for players to navigate the Kākāpō bird to safety by avoiding a myriad of obstacles which approach with increasing speed as the game advances. Players also learn about the plight of the Kākāpō bird and current conservation measures during gameplay through pop-up information boxes and quiz questions.

We focused on addressing two main research questions: (1) what is the potential effect of conservation focused mobile games on knowledge of the Kākāpō and its conservation, as well as different constructs of the theory of planned behavior (Kan & Fabrigar, 2017) (i.e., attitudes, subjective social norms and behavioral intentions) compared to a similar mobile app game with no conservation messaging and; (2) can playing a conservation focused mobile game influence players' donation behaviors to a related conservation charity when compared to a similar mobile app game with no conservation messaging.

The theory of planned behavior, a framework used to understand and predict behaviors, describes behaviors as being directly determined by behavioral intentions and under certain circumstances, perceived behavioral control (Figure 2). Behavioral intentions are in turn determined by attitudes, subjective social norms, and perceived behavioral control (Kan & Fabrigar, 2017). We also included knowledge as part of our theoretical model (Figure 2) based on the hypothesis that knowledge of the Kākāpō and its conservation could have an indirect impact on pro-environmental behavior, as shown by Liu et al. (2020). We decided to focus on the intention of players to perform four different pro-environmental behaviors which related to the Predator-Free New Zealand 2050 strategy, proposed in 2016. The strategy aims to eradicate invasive non-native mammalian predators and thus restore New Zealand's

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FIGURE 1 The digital mobile game Kākāpō Run, inspired in the "endless runner" genre requires players to help the Kākāpō avoid different types of obstacles, from rolling boulders to invasive predators, across three environments: Forest, Coastline, and City.

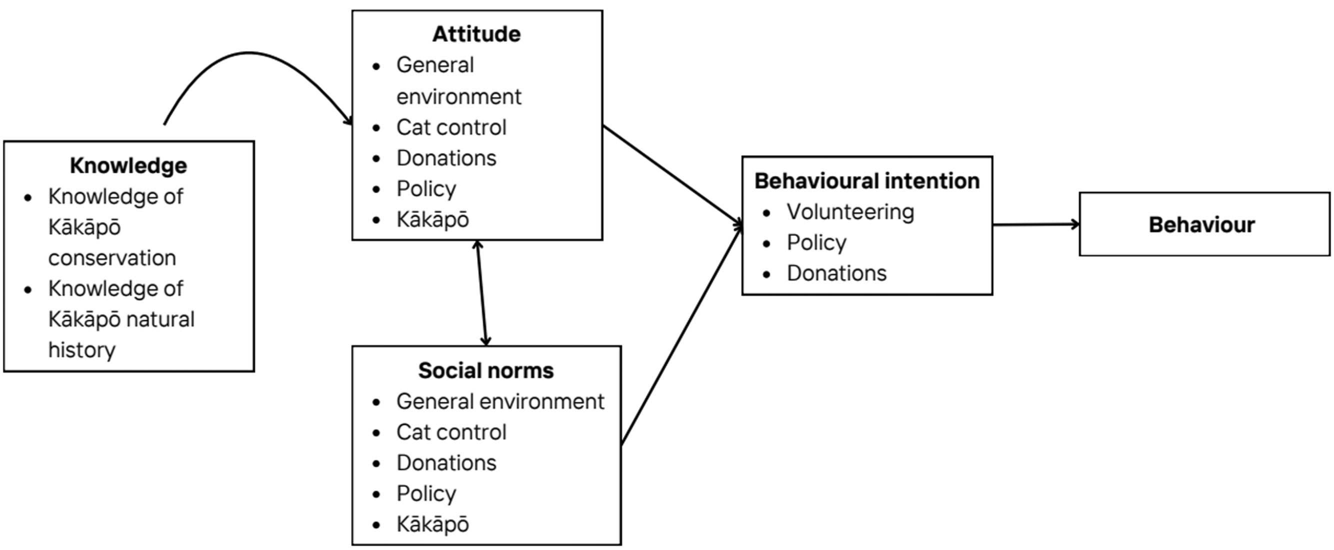

FIGURE 2 Structure and component of the theory of planned behavior (Ajzen, 1991) with knowledge as an additional component.

native biological diversity (Peltzer et al., 2019). The behaviors included the intention to donate toward, volunteer for or support a policy that aimed to remove invasive rats, stoats, and possums, from New Zealand, as well as the intention to employ measures to control their domestic cats from roaming outside.

2 | METHODS

We used a randomized controlled trial to evaluate the impact of the mobile app game "Kākāpō Run" on pro-environmental behaviors. The trial included two conditions: a treatment group (playing "Kākāpō Run" for 1 h) and a control group (playing "Subway Surfer," a digital mobile game with a similar gameplay but no conservation messaging, for 1 h) to which participants were assigned randomly, unaware of which group they belong to.

The randomized controlled trial was conducted in New Zealand, the only country where the Kākāpō occurs. Participants were recruited using a New Zealand-based research recruitment agency. Each participant was offered a cash incentive of $70 NZ. We screened participants to attempt to balance treatment and control samples across demographics (e.g., age, gender, ethnicity), as well as interest in environmental conservation and in mobile gaming. Participants that were members of an environmental organization or had watched a nature documentary in the last two weeks were classified as having an interest in the environment. Participants that had spent over three hours playing a mobile game in the last week were classified as having an interest in gaming. Only individuals with a prior interest in conservation, gaming or both were included in our final sample to better represent those likely to play the game.

Study participants were asked to complete a 10-minute questionnaire before spending the following seven days completing their assigned task (see Supporting Information Initial Survey). After the 7-day period, all participants were asked to re-take the questionnaire (Figure S1).

The questionnaire was modeled on the behavioral constructs identified within the theory of planned behavior model (Ajzen, 1991) with the addition of knowledge (see Supporting Information Initial Survey). These constructs focused on knowledge, attitudes, and social norms. Each construct (apart from knowledge) was then designed to include four sub-constructs focusing on (1) the general environment, (2) donations toward Kākāpō bird conservation, (3) the Kākāpō bird, and (4) policies to protect the Kākāpō bird. Each question was structured as statements with a 5-point Likert scale response (full questionnaire found in the Supporting Information), with some constructs being measured using multiple statements and some are measured with single statements. The theory of planned behavior was tested as a candidate framework upon which to measure the success of conservation messaging placed in games. We also included demographic questions to elicit information on the participant's background, including their formal education level, age, and gender.

To measure behavioral intentions, participants were asked how likely it would be for them to donate toward, volunteer for an NGO supporting Kākāpō conservation or support a policy that aimed to remove non-native mammalian predators from New Zealand. Answers were recorded on a 5-point Likert scale from "*highly likely*" to "*highly unlikely*." Participants were then asked if they currently owned a cat and whether they allowed the cat to roam outdoors. If the participant responded "yes" to both these questions then they were then asked whether they would consider employing one of three control measures for domestic cats, including: keeping the cat indoors; putting a bell on the cat; limiting the number of hours the cat can spend outside. These control measures were selected as they have been previously found to limit cat predatory behaviors to varying degrees (Trouwborst & Somsen, 2020).

We measured donation behaviors by asking participants at the end of the second survey if they would like to donate a portion of their incentive payment ($70 NZ) to one of two charities linked to Kākāpō conservation. To measure the retention of any donation behaviors, we re-surveyed participants six weeks following the second survey. This research was approved by the Central University Research Ethics Committee, University of Oxford (R70049/RE001).

We conducted a priori power analysis in RStudio (1.2.5019) using the package pwr (Champely, 2018), using the following parameters: alpha $= 0.05$, power $= 0.8$ and between group effect size $= 0.2$ (Cohen, 2013). Results determined that a minimum sample of 99 participants per group would be needed to detect this change. We sampled 200 participants in total, equally divided between treatment and control groups, which provides a statistical power of 0.8.

2.1 | Data analysis

We analyzed the data using R (version 4.1.0). First, we calculated participants' scores for "general environmental" attitudes, and subjective social norms by summing the Likert scale scores across the questions related to each construct. Questions that were negatively worded were reverse scored before inclusion. We coded participants' responses to the question about employing cat control measures as "1" if willing to employ these measures and "0" if not. Similarly, knowledge scores were coded as binary for each question where "1" was for a correct answer and "0" was for an incorrect answer. We used ordinal alpha to determine whether items could be analyzed jointly or had to be looked at separately (Table S1). Following van Griethuijsen et al. (2015) we used values above 0.6 as an acceptable threshold for internal consistency.

We used standardized mean difference to compare control and treatment groups at baseline, to assess comparability. The standardized mean difference scores for each variable (i.e., age, gender, education, ethnicity, gaming interest, and environmental interest; see Table S2) were compared using Cohen's d (Cohen, 2013). If a variable was found to have a large effect size (i.e, $d > 0.2$), it was added as a fixed effect in our subsequent analysis models to control for this variation.

To estimate the effect size and direction of change in questionnaire scores in the treatment group compared to the control on the above-described indicators we used Cohen's d, calculated based on the coefficients of the different regression analysis. To consider the influence of any differences at baseline, we used a series of ordinal logistic regression models (cumulative link mixed models [CLMMs]) to measure the influence of the interaction between the intervention group and pre- and post-measure on participant questionnaire scores including fixed effects (such as treatment status, age and environmental interest), and random effects such as the participants' unique ID. For this we used the packages lme4 (Bates et al., 2014) and ordinal (Christensen, 2019). As a metric of overall model fit for each CLMM model, we used the Hessian's conditional value. Any Hessian value below 10,000, signifies that the model is "well-defined" (Kartzinel et al., 2013). The exceptions were the knowledge related questions, intention to employ cat control measures and actual donation behaviors where we used binomial generalized linear mixed models (GLMMs) as the response variables were binary. For this we used the package lme4 (Bates et al., 2014).

We conducted two GLMMs on participants' actual donation behaviors: one to investigate the donation behaviors across the control and treatment groups in the second survey; and one to investigate the retention of donation behaviors between the second survey and 6-week follow-up question which included an interaction between the group and time. For this we used the packages lme4 (Bates et al., 2014).

3 | RESULTS

3.1 | Sample

The demographic characteristics of our sample is described in Table S2. When comparing the two groups using standard mean differences, previous environmental interest and age were both found to have a bigger than small standardized mean difference (i.e, $d > 0.2$), as defined by Cohen (2013), between the control and treatment groups (Table S2). These variables were therefore included as fixed effects alongside the random effect of participant ID within our mixed models.

3.2 | Survey results

Based on ordinal alpha scores, items within the knowledge construct were analyzed independently whereas items within the attitude and social norm constructs were each analyzed together (Table S1). Participants across both the control and treatment groups were found to have pre-existing high levels of knowledge across all questions about the Kākāpō bird and its conservation (Table 1). We found participants in both treatment and control groups had a mean score above 4 out of 5 on the pre-intervention survey questions relating to knowledge (Q1–5). This trend was also noted across participant's pre-intervention scores for general environmental attitudes (participants in both the control and treatment group had a mean score of above 12 out of 15 points which indicates a high positive attitude toward the environment; Figure S2).

When accounting for the baseline differences between treatment and control groups, the ordinal regression model (Table S3) found that three out of five of the questions relating to participants knowledge were found to be significantly positively influenced by the intervention (Question 2: GLMM model: $\beta = 0.23$, SE $= 0.062$, $p < 0.05$; Question 3: $\beta = 0.21$, SE $= 0.068$, $p < 0.05$; Question 4: $\beta = 0.22$, SE $= 0.062$, $p < 0.05$). We also found that participant's attitudes toward policy was significantly positively influenced by the intervention ($\beta = 1.312$, SE $= 0.525$, $p < 0.05$) (Figure S2; Table S4). There was no significant influence of the treatment intervention across any sub-constructs of subjective social norms (CLMM model: general environment: $\beta = 0.146$ [SE $= 0.443$], donation: $\beta = -0.115$ [SE $= 0.430$]; Kakapo: $\beta = 0.365$ [SE $= 0.542$]; policy: $\beta = 0.146$ [SE $= 0.443$]) (Figures 3 and S4).

The models also suggest that previous environmental interest had a significant positive influence on a number of sub-constructs across attitudes and social norms, including attitudes and social norms surrounding donating toward invasive species control (CLMM: $\beta = 1.251$ [SE $= 0.484$], $p = 0.05$; $\beta = 1.301$ [SE $= 0.388$], $p < 0.05$, respectively); and attitudes and social norms surrounding invasive species control (CLMM: $\beta = 1.833$ [SE $= 0.422$], $p < 0.05$; $\beta = 1.121$ [SE $= 0.499$], $p < 0.05$, respectively). In addition, our models found that being in the age category 46–55 was associated with significantly higher knowledge Q4 scores (linear mixed effects model: $\beta = 0.198$ [SE $= 0.074$], $p < 0.05$), and being in the 36–45 group was also associated with significantly higher general environment and policy subjective social norms scores (CLMM: $\beta = -1.110$ [SE $= 0.002$], $p < 0.05$; $\beta = -1.11$ [SE $= 0.0002$], $p < 0.05$, respectively). Overall, pseudo R^2 metrics were low (Table S4).

3.3 | Behavioral intentions

We found no evidence across both the control and treatment groups of respondents changing their intention to donate to invasive species control post the intervention (Figures 4 and 5; Table S4). However, intentions to vote for invasive species control policies and volunteer to

TABLE 1 The mean scores of participants in the treatment ($n = 100$) and control ($n = 100$) groups both pre- and post- the mobile app game interventions including standard errors in brackets.

		Treatment		Control	
		Pre	Post	Pre	Post
Knowledge	Q1	0.99 (0.01)	1.00 (0.0001)	0.99 (0.01)	0.99 (0.01)
	Q2	0.68 (0.05)	1.00 (0.0001)	0.65 (0.05)	0.24 (0.04)
	Q3	0.67 (0.05)	0.97 (0.02)	0.71 (0.05)	0.80 (0.04)
	Q4	0.71 (0.05)	0.99 (0.01)	0.70 (0.04)	0.70 (0.04)
	Q5	1.49 (0.05)	1.62 (0.05)	1.48 (0.06)	1.50 (0.16)
Attitudes	General environment	12.51 (0.17)	12.41 (0.16)	12.31 (0.17)	12.10 (0.16)
	Donations	4.07 (0.07)	4.13 (0.07)	4.07 (0.07)	3.99 (0.07)
	Kākāpō	3.82 (0.08)	4.00 (0.06)	3.82 (0.08)	3.88 (0.07)
	Policy	4.36 (0.07)	4.44 (0.06)	4.40 (0.07)	4.26 (0.07)
Social Norms	General environment	8.81 (0.21)	8.70 (0.22)	8.89 (0.22)	8.65 (0.21)
	Donations	3.75 (0.08)	3.65 (0.08)	3.74 (0.08)	3.99 (0.08)
	Kākāpō	2.96 (0.07)	3.00 (0.08)	2.87 (0.09)	2.93 (0.08)
	Policy	3.80 (0.06)	3.86 (0.05)	3.76 (0.06)	3.74 (0.06)
Behavioral Intentions	Cat control	4.14 (0.08)	4.20 (0.06)	4.18 (0.07)	4.07 (0.07)

invasive species control efforts, were both positively influenced by exposure to the treatment intervention (Figure 5; CLMM: $\beta = 0.048$ [SE $= 0.0001$], $p < 0.05$; $\beta = 1.448$ [SE $= 0.430$], $p < 0.05$, respectively).

Of the total 200 participants included in this study, 35 in the control group and 36 in the treatment group owned a cat that they allowed to roam freely outdoors. Of these individuals, we found that a higher proportion of those in the treatment group had the intention to employ cat control measures after the intervention than those in the control group (Figure 5). The GLMM model found the intention to employ cat control measures was significantly influenced by the interaction between the intervention group and the pre–post-stage of the intervention (GLMM: $\beta = 2.592$ [SE $= 1.142$], $p = 0.02$) (Table S5). The effect size (Cohen's d) for this change was 1.22, CI [0.32, 2.62]. Overall, pseudo R^2 metrics were low (Table S4).

3.4 | Donation behaviors

We found that across both the control and treatment groups most respondents did not donate part of their incentive to any of the charities related to Kākāpō conservation, with just 19% of participants in the treatment group and 13% of participants in the control group choosing to donate their incentives at the end of the second survey. Of those that chose to donate, the donation amounts ranged from \$1 NZ to \$70 NZ, with an mean donation amount of \$20.26 NZ in the treatment group and \$12.82 NZ in the control. A GLMM model found that donation behaviors were not significantly influenced by placement in either the control or treatment intervention group ($\beta = 1.757$ [SE $= 15.902$], $p = 0.912$), with a Cohen's d of 0.55, CI [−0.26, 1.42] (Table S7).

The number of respondents to the follow-up questionnaire was 195 (97 in the treatment group and 98 in the control group). Results from this follow-up found that more participants were willing to donate to one of the two charities related to Kākāpō conservation six weeks post the intervention than immediately after the intervention. We found that 40% of participants in the treatment group ($n = 39$) and 36% of participants in the control group ($n = 35$) were willing to donate six weeks after the intervention.

However, results from the GLMM model found that moving from immediately after to 6 weeks following the intervention did not significantly influence participants' donation behaviors ($\beta = -0.280$ [SE $= 0.498$], $p = 0.575$) (Table S8).

4 | DISCUSSION

Mobile games are increasingly dominating the digital landscape. Yet, we know very little about their potential to drive pro-social outcomes, not only in an environmental context but even more broadly (Boyle et al., 2016). We document the impacts of the mobile game Kākāpō Run,

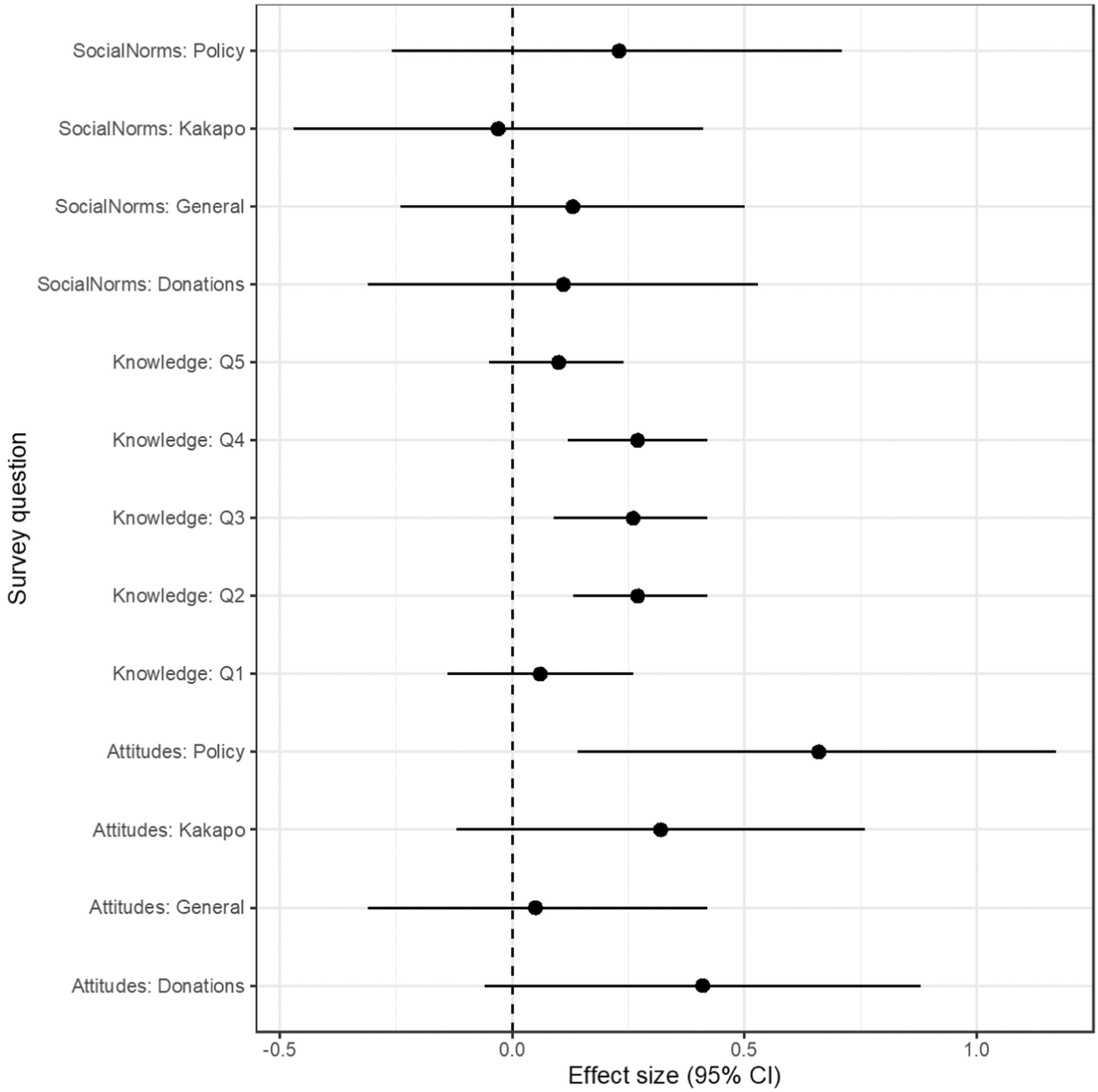

FIGURE 3 Effect size (Cohen's d) of participants' questionnaire scores increasing post exposure to the treatment intervention for each construct assessed. The dashed line indicated no change. Error bars represent 95% confidence intervals.

in what is one of the first experimental impact evaluations of a mobile digital game focused on influencing pro-environmental behaviors (Boyle et al., 2016). We found that the game was able to drive not only changes in knowledge of the Kākāpō bird, Kākāpō conservation and associated measures toward Kākāpō preservation, and attitudes toward the environment, but also in behavioral intention such as the willingness to volunteer to remove predators from their own backyard and to manage house cats more effectively. This highlights that mobile games have the potential to play an important role in influencing human behavior in a biodiversity conservation context. However, none of the models had high explanatory power and that potentially affected the significance of the results that we present here. Overall, we had low pseudo R^2 values for all our models. Our models had low explanatory power suggesting there are other factors driving the variance in the data.

5 | CAN GAMES DRIVE CHANGE?

5.1 | Knowledge

Knowledge is perhaps the most frequently measured indicator in the context of the impact evaluation of behavioral interventions focused on pro-environmental behaviors and of serious games (Boyle et al., 2016; Gralton et al., 2004). Knowledge had multiple questions with positive significant effect sizes (Figure 3), even though baseline levels were generally already high. The magnitude change measured was in line with that

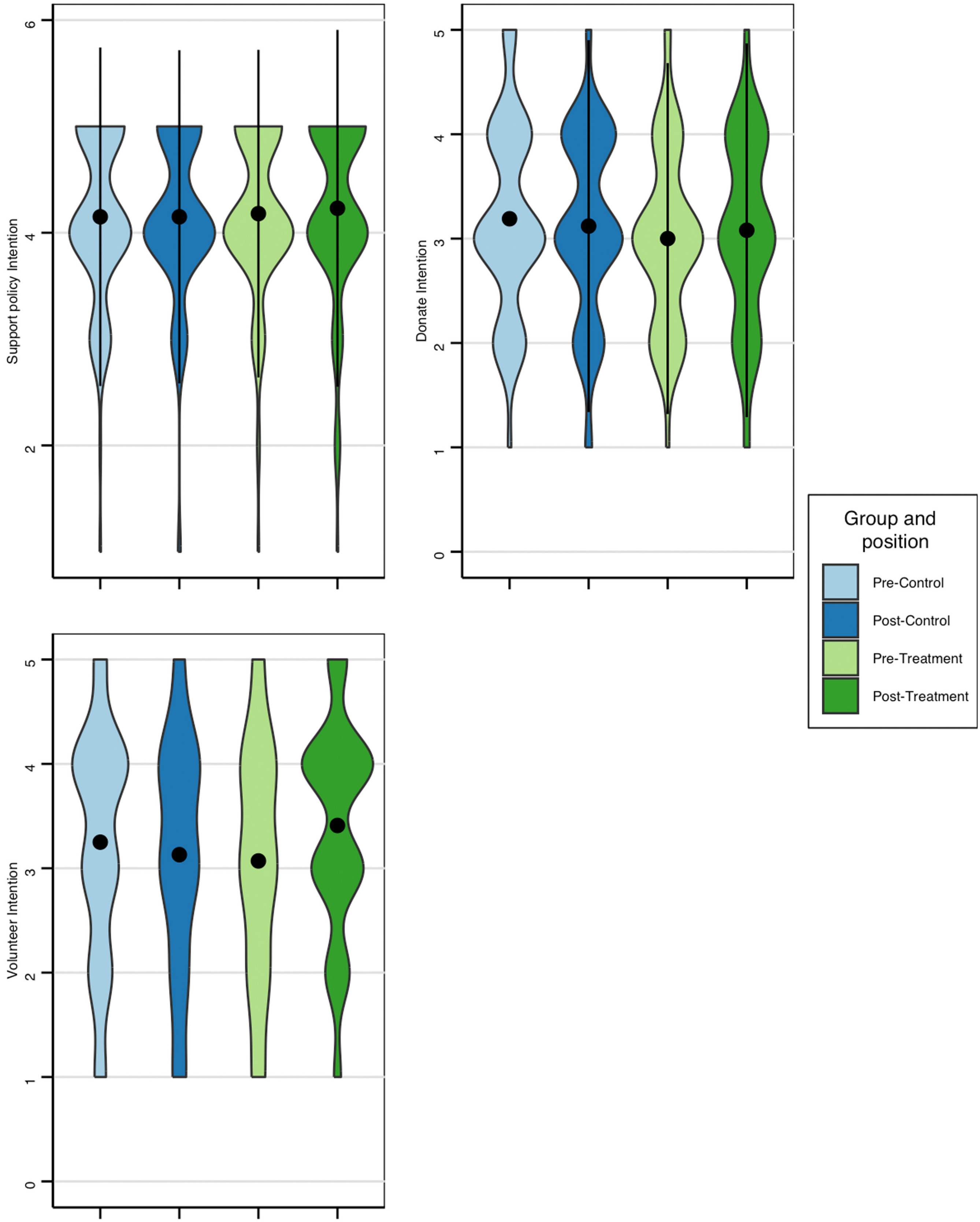

FIGURE 4 Violin plots showing the range of survey scores across the different behavioral intentions measured within the survey both before and after exposure to the interventions in the control and treatment groups, $N = 200$. The whiskers are the upper and lower quartile values and dots are the outlier data points.

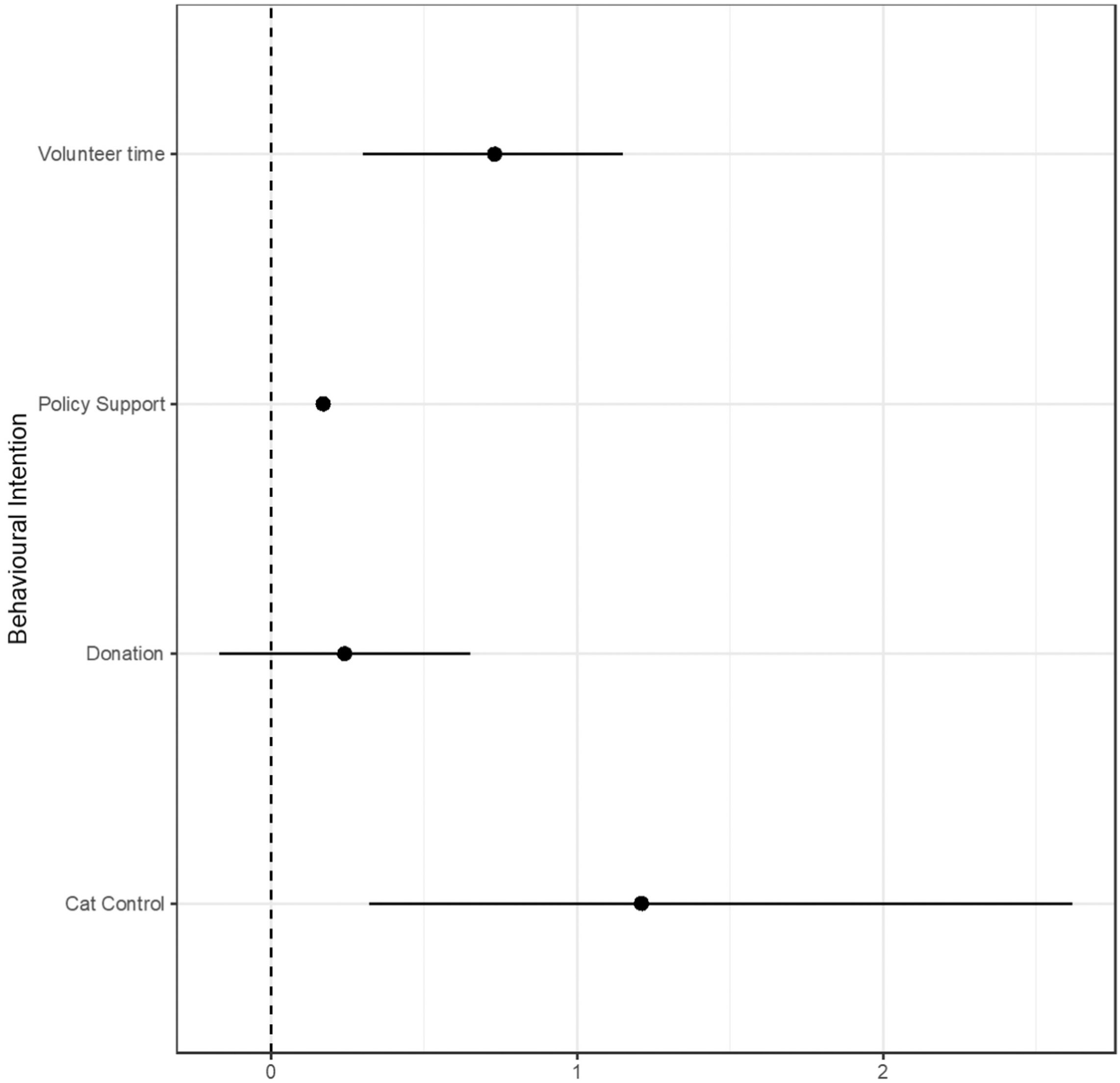

FIGURE 5 Effect size (Cohen's *d*) of participants' questionnaire scores increasing post-soexposure to the treatment intervention for each behavioral intention assessed. Error bars represent 95% confidence intervals.

documented across serious games more broadly (Boyle et al., 2016). However, it is important to be aware that this was also the indicator where change is most often observed, and that the relationship between increased knowledge and behavior change is often weak at best (Boyle et al., 2016; Kollmuss & Agyeman, 2002). Furthermore, given that the participants were so highly knowledgeable about Kākāpō conservation issues, our results may not be generalizable to other conservation contexts in terms of knowledge gain post-intervention.

5.2 | Theory of planned behavior constructs

In terms of attitudes, the observed change in three out of four attitudinal constructs examined (Figure 3) aligns Kākāpō Run with a meta-analysis of serious games by Wouters et al. (2013), which found positive effect size, but no statistical significance. That said, it is worth noting that the attitude variable focused on policy was positive and statistically significant, which is meaningful in terms of harnessing support for the Predator-Free New Zealand 2050 strategy.

Regarding social norms, the lack of change was somewhat expected, given that there was no multiplayer mode or indeed any opportunities for players to interact with each other within the game. The introduction of a social element to the gameplay is an area that a future version of Kākāpō Run could explore. While there is very limited research on the ability of serious games, environmental or otherwise, to shift social norms (Boyle et al., 2016), it has been suggested that some types of serious games could influence norms

(Hildmann et al., 2009). With the emergence of ever more immersive games such as massive multiplayer online role-playing games, which have been suggested to play an important role in shaping the social life of some gamers (Weissman, 2017), the potential for games to impact social norms may increase.

We found positive impact on behavioral intentions to volunteer (Figure 5), an important outcome given how much volunteers contribute to conservation in general and to Kākāpō conservation in particular (Chandler et al., 2017; Jansen, 2005; Lorimer, 2009; Towns et al., 2012). We also found an increase in intention of players to perform pro-environmental behaviors related to the Predator-Free New Zealand 2050 strategy, proposed in 2016 (Peltzer et al., 2019). This could be a game-changer for birds such as the Kākāpō, which have invasive predators as their key threat, and public support for this policy is a key part of guaranteeing that it will be implemented (Peltzer et al., 2019).

The change in behavioral intention around pet cat management (Figure 5) is encouraging given the importance of that issue for the conservation of many endemic birds in New Zealand (Rouco et al., 2017). While this is not immediately relevant for Kākāpō conservation given that the species lives in offshore islands only distant from where the participants of the study were based, this spillover result highlights the role of the Kākāpō as a conservation flagship species. Yet, this change in behavior was somewhat unexpected given that no cats were featured in the game. While the error in the estimated effect size means that our result should be seen as hypothesis generating as opposed to hypothesis testing, the scalability of mobile games means this is no doubt something that deserves further research.

There was also no change in terms of willingness to donate or actual donations, even if the respondents were chosen to come from New Zealand, and so have at least theoretically more proximity with the target of the intervention. There was potentially a change in donations, since there were donators of a higher value in the treatment groups, though there was too much variance to be significant in these groups. This no doubt presents a challenge to conservation gaming as biodiversity is only going to benefit if actual behavior is influenced. Donation elicitation is not an area that many serious games have focused on (Boyle et al., 2016), although given how commonplace in-app financial transactions have become this is an area where further exploration is warranted. This suggests that developers of serious games may face a trade-off when prioritizing between a game environment that is focused on user experience and key messages, which is more likely to be conducive to change versus one focused on the adverts and in-app purchases that are common in commercial games, which is more likely to result in revenue.

It is true that having donations as the sole behavioral indicator is limiting, as donation behavior can be influenced by the content and characteristics of the respondent and there are many other behaviors that are relevant to biodiversity. This may be particularly the case for those engaged in a paid activity like marketing research, where participants may only partake in research to get more money. If this is the case, we can assume that they need extra income and so would be less likely to give it away. In that context, it is key to use more diverse pro-environmental behavioral indicators. Still, donations are one of the simplest behaviors to measure directly (i.e., without reliance on self-reporting) in an experimental setting. In our analysis we also noted that a small number of ordinal regression models violated the proportional assumption (Table S6) but judged that given the trade-offs between these and other model types, ordinal regression was still the best available option for this analysis (Harrell, 2015).

6 | BARRIERS TO GAMING IMPACT

Despite their potential, there are several barriers to the use of games, even before considering interventions. Despite the global popularity of digital games, this approach is still limited by its reliance on an infrastructure that is simply not available to many people around the world, creating a digital divide between, for example, high and low income countries and groups, but also between age and gender groups (International Telecommunication Union, 2023). This issue is even more acute when games rely on cutting-edge technology, such as virtual or augmented reality, which relies on higher digital literacy and top performing hardware (Dunn et al., 2021).

Finding the balance between game playability, fun and solid learning design that aligns learning outcomes is a critical challenge for effective serious game design. In the case of Kākāpō Run, the game designers aimed to explicitly make the game entertaining, with the knowledge transmission aspect, including the pop-up fact boxes and quiz questions, being secondary. In a review of game evaluations, Boyle et al. (2016) found that enjoyable game aspects, and particularly those that lead to positive impacts, were consistently earning points, finding rare items, and fast loading times, as opposed to more learning-focused aspects. In the game's design presented here, a points (in this case feathers) collecting aspect was created, and loading times were adjusted to ensure a quick experience.

With more than 110,000 players in its first year, Kākāpō Run, has enjoyed a reasonable level of popularity compared to a mean app download number of 27,000 across all iOS and Android devices (Curry, 2022). Yet, there are several barriers to extrapolating our results to this larger group. The first is that only about 2% of these were from New Zealand. Most downloads by far came from India, with the game organically becoming of the Android store's most downloaded games in Asia for several weeks. The second is that with a playtime of about 30 min across the first year after launch, the average player plays less than the time used for this experiment as therefore the effect may be less pronounced.

Furthermore, there are limitations to our use of the theory of planned behavior as a framework for a more rigorous impact evaluation on the role of gaming on promoting conservation attitudes and behaviors, given its generalist focus. These limitations manifest in the absence of gaming-specific sub-constructs, such as "flow," the state of being fully engaged in an activity, that could have shed new insight into how game design impacts outcomes (Hamari & Koivisto, 2014). This is why in other work (Thomas-Walters & Veríssimo, 2022) we have use the theory-based framework for gamification research (Treiblmaier et al., 2018), which brings together multiple behavioral models relevant to gaming research. In this case, the use of the theory of planned behavior is further limited by a lapse in survey design that did not allow for the measurement of perceived behavioral control. To avoid this we have used only already validated scales, such as Alisat's Environmental Action Scale or Milfont's Environmental Attitudes Inventory (Thomas-Walters & Veríssimo, 2022) in other work on this topic.

7 | TOWARD CONSERVATION GAME SCIENCE

Recognizing the potential of digital games to yield positive conservation outcomes requires acknowledging the need for a better understanding of how we can optimize games to improve learning and behavioral outcomes. This can only be achieved through more extensive and higher quality research into game design, development and evaluation, a realization that led to the emergence of the field of "game science" (de Freitas, 2018). In line with the findings of Clark et al. (2016) regarding the broader landscape of games, the literature on digital games and biodiversity has so far mainly remained conceptual, describing gameplay and hypothesizing about potential positive outcomes but without empirical investigation of these claims (e.g., Colleton et al., 2016; Phipps et al., 2016; Salvador & Cunha, 2016). If conservationists are to leverage the potential of digital games, then two fundamental research axes will need to be focused on: (1) understanding user experience and (2) evaluating the impact of playing games.

We need to understand the complexities of designing and disseminating games that are designed to implement social change, something that is recognized in the serious games literature but has had limited research in the biodiversity conservation context. Mobile gaming is very competitive with dozens of games being released every day, all competing for user attention. While one of the big advantages of mobile games may be the ability to reach non-traditional audiences without an initial motivation to engage with content linked to nature, this only happens if conservation games are attractive to these users and that requires a detailed understanding of gamer preferences and experiences when playing. This parallels work done in other areas of conservation marketing, where audience research is acknowledged as a key element (Veríssimo et al., 2020).

In terms of evaluation, it is key to acknowledge that we do not yet have an evidence base that allows for broader conclusions around the use of mobile games for biodiversity conservation. This is despite the broadly positive outlook conservationists have of them (e.g., Fisher et al., 2021). Of particular interest would be an investigation of the partnerships between established games and conservation organizations, which have been celebrated as impactful (Sandbrook et al., 2015) but for which little to no information exists. Theoretically, capitalizing on conservation impact in existing games may be more cost-effective than making an original educational game, particularly if the budget available is small. Yet, the absence of information leaves the issue open to speculation.

This impact evaluation of Kākāpō Run showcases both the potential of mobile games in the context of conservation outreach and marketing, and the importance of rigorous impact evaluation. We call for other conservationists engaged in designing and implementing digital interventions to approach game design and evaluation in a more research-centered way to help develop an evidence base around the intended and unintended consequences of game playing and both positive and negative impacts.

In the work presented here, it was at times challenging to retrofit impact assessment of an environmental game to the pre-existing framework of the theory of planned behavior. A lack of significant, positive consequences from playing Kākāpō Run does not dispel existing robust evidence for the impacts of playing serious games. These results highlight the ongoing search for the optimization of best practice impact evaluation for the growing genre of environmental games. This mainstreaming of gaming science across conservation will be critical to allow for mobile games to realize their true potential as a leading communication channel for behavioral change in biodiversity conservation.

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DATA AVAILABILITY STATEMENT

Data is available from the authors upon request.

ORCID

Diogo Veríssimo https://orcid.org/0000-0002-3519-6782

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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